SOLUTION

BOTH VERSIONS OF THE EXAM ARE IN THIS FILE

CADET _____ SECTION _____TIME OF DEPARTURE _____

DEPARTMENT OF CHEMICAL & BIOLOGICAL SCIENCE & ENGINEERING

CH365 2026-2027
WRITTEN PARTIAL REVIEW II
15 October 2026, A-Hour

TEXT: Smith, Van Ness, Abbott & Swihart
SCOPE: Lessons 10-20
TIME: 55 Minutes

References Permitted: Open notes, book, internet, Mathematica, and Excel.

INSTRUCTIONS

1. Do not mark this exam or open it until "begin work" is given.
2. You have 55 minutes to complete the exam.
3. There are 3 problems on 3 pages in this exam (not including the cover page). Write your name on the top of each sheet.
4. Solve the problems in the space provided and in Mathematica. Show all work to receive full credit.
5. Save Mathematica files on your desktop and upload to CANVAS when finished.

(TOTAL WEIGHT: 200 POINTS)

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
A	50	
B	70	
C	80	
TOTAL CUT		
TOTAL GRADE	200	

Cadet: **SOLUTION**

Problem:	Weight:
A	50

Styrene ($C_8H_8(g)$) is fed to a combustion reactor at 1.0 bar with 150% excess air and at the temperature of the reactor.

(a) Balance the reaction on a basis of one mole of styrene in the feed.

(b) Calculate the heat of reaction at 25°C and 1.0 bar in J/mol on a basis of one mole of styrene in the feed.

(c) The reactor is isothermal at 1.0 bar with a heat duty of -4,261.203 kJ/(mol styrene). Determine the temperature of the reactor in °C.

The following table contains ideal gas heat capacity polynomial coefficients and standard heats of formation at 298 K from Appendix C:

species	a	$b\times10^3$	$c\times10^6$	$d\times10^{-5}$	$\Delta H^0_{f,298}$, J/mol
$C_8H_8(g)$	2.050	50.192	-16.662	0.000	147,360
$O_2(g)$	3.639	0.506	0.000	-0.227	0
$N_2(g)$	3.280	0.593	0.000	0.040	0
$CO_2(g)$	5.457	1.045	0.000	-1.157	-393,509
$H_2O(g)$	3.470	1.450	0.000	0.121	-241,818

Solution, part (a):

Balanced reaction in 150% excess air:

$$C_8H_8(g) + 25\ O_2(g) + 94.05\ N_2(g) \rightarrow 8\ CO_2(g) + 4\ H_2O(g) + 15\ O_2(g) + 94.05\ N_2(g)\ \text{//ANS}$$

Solution of parts (b) and (c) in Mathematica are shown on the following page.

Solution, part (b):

```
(*List of stoichiometric coefficients*)
(*styrene, oxygen, nitrogen, carbon dioxide, water, oxygen, nitrogen*)
ν = {-1, -25, -94.05, 8, 4, 15, 94.05};

(*List of standard Heats of Formation*)
ΔHf298 = {147 360, 0, 0, -393 509, -241 818, 0, 0};

(*Standard Heat of Reaction at 298 K*)
ΔH298 = ν.ΔHf298 (*J/mol*)
```

$-4.262\,704 \times 10^{6}$

ANS, J per mole of styrene

Solution, part (c):

```
(*Heat Capacity polynomial coefficients, Given, from Table C.1:*)
a = {  2.050,  3.639, 3.280,  5.457, 3.470,  3.639, 3.280};
b = {  50.192,  0.506, 0.593,  1.045, 1.450,  0.506, 0.593} * 10^-3;
c = {-16.662,  0.000, 0.000,  0.000, 0.000,  0.000, 0.000} * 10^-6;
d = {  0.000, -0.227, 0.040, -1.157, 0.121, -0.227, 0.040} * 10^5;

Δa = ν.a;
Δb = ν.b;
Δc = ν.c;
Δd = ν.d;

(*Temperatures, K*)
(*T=unknown*)
T0 = 25 + 273.15;

(*Gas constant*)
R = 8.314; (* J/(mol*K) *)

(*Equation 4.21 for MDCPH*)
MDCPH = Δa + Δb/2 * (T + T0) + Δc/3 * (T^2 + T0^2 + T * T0) + Δd/(T * T0);

(*Equation 4.22 for corrected heat of reaction*)
(*Gives molar heat of reaction in J/mol*)
(*ΔH573=-4.261*10^6 J/mol given*)
SolveValues[-4.261 203 * 10^6 == ΔH298 + MDCPH * R * (T - T0), T] - 273.15

{-87.60636, 167.0947, 300.0462, 2 227.181}
```

ANS, °C (most likely of the four possible correct answers)

Cadet: **SOLUTION**

Problem:	Weight:
B	70

An engineering plan for a polypropylene plant includes a pipeline carrying propylene (C_3H_6) at a volumetric flow rate of 83.201 MMSCFD (million standard cubic feet per day). Standard conditions are 519.67°R and 14.696 psia. At these conditions, the molar volume of C_3H_6 is 373.70 ft^3/lbmol.

(a) Calculate the molar flow rate of C_3H_6 in lbmol/sec.

(b) Use the Soave-Redlich-Kong (SRK) equation of state (equations 3.41, 3.44, and 3.45), constants in Table 3.1, and gas constant R = 10.73 $ft^3 \cdot psia/(lbmol \cdot °R)$ to determine the isothermal reversible compression work required to reduce the molar volume of C_3H_6 by 10%. Report the work in units of hp.

From Appendix B, the critical temperature, pressure, and acentric factor for propylene are T_C = 658.08 °R, P_C = 676.60 psia, and ω = 0.140.

Solution, part (a):

$$\frac{83{,}201{,}000\,\text{ft}^3}{\text{day}} \cdot \frac{1\,\text{day}}{24\,\text{h}} \cdot \frac{1\,\text{h}}{3600\,\text{s}} \cdot \frac{1\,\text{lbmol}}{373.70\,\text{ft}^3} = \underbrace{2.5769\ \frac{\text{lbmol}}{\text{s}}}_{\text{ANS}}$$

Solution, part (b):

```
T = 519.67; (*Rankine, given*)
R = 10.73; (* ft^3*psia/(lbmol*R), given*)
ω = 0.140; (*App. B, p.665*)
Tc = 658.08; (*Rankine, given*)
Pc = 676.60; (*psia, given*)
Tr1 = T / Tc; (*reduced temperature*)
(*Table 3.1, p.100*)
σ = 1;
ϵ = 0;
Ω = 0.08664;
Ψ = 0.42748;
α = (1 + (0.480 + 1.574 * ω - 0.176 * ω^2) * (1 - √Tr1))^2;
(*Equations 3.41, 3.44, and 3.45*)
b = Ω * R * Tc / Pc;
a = Ψ * α * R^2 * Tc^2 / Pc;
P = R * T / (V - b) - a / ((V + ϵ * b) * (V + σ * b));
```

$$W = -\int_{373.70}^{0.90*373.70} P\,dV \quad (*\text{eq 1.4, units } \tfrac{psia*ft^3}{lbmol}*)$$

578.1504

$$\frac{W*psia*ft^3}{lbmol} * \frac{\frac{lbf}{in^2}}{psia} * \left(\frac{12\ in}{ft}\right)^2 * \frac{2.576\,865\ lbmol}{s} * \frac{1.34102\ hp}{737.562\ \frac{ft*lbf}{s}}$$

390.0602 hp

ANS

Cadet: **SOLUTION**

Problem:	Weight:
C	80

0.200 kmol of an ideal gas in a closed system is changed from an initial state of 1.5 m^3, 5.0 bar and 451.0 K to a final state of 5.0 m^3, 1.5 bar, and 451.0 K by a two-step process consisting of an adiabatic expansion followed by heating at constant pressure.

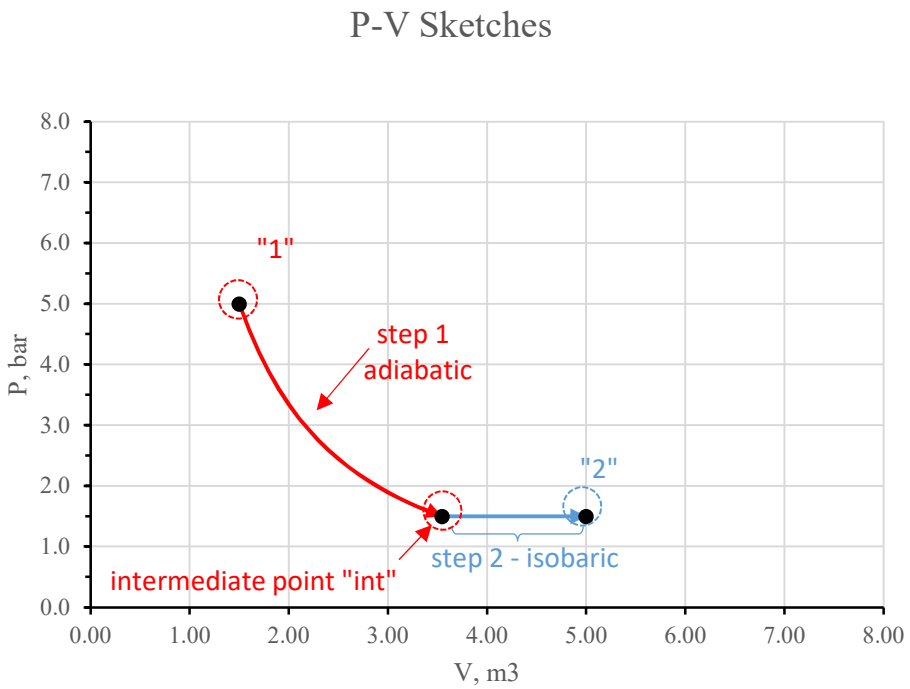
(a) Sketch the process path on the PV axes below.

(b) Calculate Q, W, ΔU and ΔH for the overall process in units of kJ.

(c) Calculate the intermediate temperature after step 1.

$$R = 8.314\ J/(mol\cdot K) = 0.08314\ (bar\cdot m^3)/(kmol\cdot K)$$
$$C_P = (7/2)\cdot R, \text{ and}$$
$$C_V = (5/2)\cdot R$$

Solution, part (a):



Solution, part (b):

The overall process is isothermal (T_2=T_1), so $\Delta H_{Overall} = \Delta U_{Overall} = 0\,kJ$ ANS

Intermediate Volume (V_{int}):

Use equation 3.23c, with the intermediate pressure $P_{int} = P_2$ (Step 2 at constant P).

Equation 3.23c needs $\gamma = C_P/C_V = (7R/2)/(5R/2) = (7/5) = 1.4$ from equation 3.24.

$$P_{int}V_{int}^{\gamma} = P_1V_1^{\gamma} \Rightarrow V_{int} = \sqrt[\gamma]{\frac{P_1}{P_{int}}V_1^{\gamma}} = \sqrt[1.4]{\frac{5.0\,bar}{1.5\,bar}\left(1.5\,m^3\right)^{1.4}} = 3.544672\,m^3$$

Work:

Step 1, adiabatic, equation 3.26:

$$W_{Step1} = \frac{P_{int}\cdot V_{int} - P_1\cdot V_1}{\gamma - 1} = \frac{1.5\,bar\cdot 3.544672\,1m^3 - 5.0\,bar\cdot 1.5\,m^3}{1.4-1} = -5.457479\,bar\cdot m^3$$

Step 2, constant pressure, work = -PΔV:

$$W_{Step2} = -P\cdot\Delta V = -P\cdot\left(V_2 - V_{int}\right) = -1.5\cdot\left(5.0\,m^3 - 3.544672\,m^3\right) = -2.18299\,bar\cdot m^3.$$

Overall:

$$W_{Overall} = W_{Step1} + W_{Step2} = -5.45748\,bar\cdot m^3 + \left(-2.18299\right)bar\cdot m^3 = -7.64047\,bar\cdot m^3$$

Convert the units to kJ:

$$W_{Overall} = -7.640471\,bar\cdot m^3\left(\frac{1\,J}{10^{-5}\,bar\cdot m^3}\right)\cdot\left(\frac{1\,kJ}{1000\,J}\right) = \underline{\underline{-764.047\ kJ}}_{ANS}$$

Heat:

$$\Delta U_{Overall} = 0 \quad \Rightarrow \quad Q_{Overall} = -W_{Overall} = \underline{\underline{764.047\,kJ}}_{ANS}$$

Solution, part (c):

$$PV = nRT \Rightarrow T_{int} = \frac{P_{int}\cdot V_{int}}{n\cdot R} = \frac{1.5\,bar\cdot 3.544672\,m^3}{0.200\,kmol\cdot\left(0.08314\,\frac{bar\cdot m^3}{kmol\cdot K}\right)} = \underline{\underline{319.8\,K}}_{ANS}$$

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C	80	
TOTAL CUT		
TOTAL GRADE	200	

Cadet: **SOLUTION**

Problem:	Weight:
A	50

n-Octane ($C_8H_{18}(g)$) is fed to a combustion reactor at 1.0 bar with 150% excess air and at the temperature of the reactor.

(a) Balance the reaction on a basis of one mole of n-octane in the feed.

(b) Calculate the heat of reaction at 25°C and 1.0 bar in J/mol on a basis of one mole of n-octane in the feed.

(c) The reactor is isothermal at 1.0 bar with a heat duty of -5,110.298 kJ/(mol n-octane). Determine the temperature of the reactor in °C.

The following table contains ideal gas heat capacity polynomial coefficients and standard heats of formation at 298 K from Appendix C:

species	a	$b\times10^3$	$c\times10^6$	$d\times10^{-5}$	$\Delta H^0_{f,298}$, J/mol
$C_8H_{18}(g)$	4.108	70.567	-22.208	0.000	-208,750
$O_2(g)$	3.639	0.506	0.000	-0.227	0
$N_2(g)$	3.280	0.593	0.000	0.040	0
$CO_2(g)$	5.457	1.045	0.000	-1.157	-393,509
$H_2O(g)$	3.470	1.450	0.000	0.121	-241,818

Solution, part (a):

Balanced reaction in 100% excess air:

$$C_8H_{18}(g) + 18.75\ O_2(g) + 70.54\ N_2(g) \rightarrow 8\ CO_2\ (g) + 9\ H_2O(g) + 6.25\ O_2(g) + 70.54\ N_2(g)$$

Solution of parts (b) and (c) in Mathematica are shown on the following page.

Solution, part (b):

```
(*List of stoichiometric coefficients*)
(*n-octane, oxygen, nitrogen, carbon dioxide, water, oxygen, nitrogen*)
ν = {-1, -18.75, -70.54, 8, 9, 6.25, 70.54};

(*List of standard Heats of Formation*)
ΔHf298 = {-208 750, 0, 0, -393 509, -241 818, 0, 0};

(*Standard Heat of Reaction at 298 K*)
ΔH298 = ν.ΔHf298 (*J/mol*)
```

$-5.115\,684 \times 10^{6}$

ANS, J per mole of n-octane

Solution, part (c):

```
(*Heat Capacity polynomial coefficients, Given, from Table C.1:*)
a = { 4.108,  3.639, 3.280,  5.457, 3.470,  3.639, 3.280};
b = { 70.567,  0.506, 0.593,  1.045, 1.450,  0.506, 0.593} * 10^-3;
c = { -22.208,  0.000, 0.000,  0.000, 0.000,  0.000, 0.000} * 10^-6;
d = {  0.000, -0.227, 0.040, -1.157, 0.121, -0.227, 0.040} * 10^5;

Δa = ν.a;
Δb = ν.b;
Δc = ν.c;
Δd = ν.d;

(*Temperatures, K*)
(*T=unknown*)
T0 = 25 + 273.15;

(*Gas constant*)
R = 8.314; (* J/(mol*K) *)

(*Equation 4.21 for MDCPH*)
MDCPH = Δa + Δb/2 * (T + T0) + Δc/3 * (T^2 + T0^2 + T * T0) + Δd/(T * T0);

(*Equation 4.22 for corrected heat of reaction*)
(*Gives molar heat of reaction in J/mol*)
(*ΔH573=-5.110298*10^6*)
SolveValues[-5.110 298 * 10^6 == ΔH298 + MDCPH * R * (T - T0), T] - 273.15

{-176.4949, 232.017, 300.0363, 2 299.275}
```

ANS, °C (most likely of the four possible correct answers)

Cadet: **SOLUTION**

Problem:	Weight:
B	70

An engineering plan for a liquified natural gas plant includes a pipeline carrying natural gas at a volumetric flow rate of 56.219 MMSCFD (million standard cubic feet per day). Standard conditions are 519.67 °R and 14.696 psia. At these conditions, the molar volume of natural gas is 376.42 ft^3/lbmol.

(a) Calculate the molar flow rate of natural gas in lbmol/sec.

(b) Use the Soave-Redlich-Kong (RK) equation of state (equations 3.41, 3.44, and 3.45), constants in Table 3.1, and gas constant R = 10.73 $ft^3 \cdot psia/(lbmol \cdot °R)$ to determine the isothermal reversible compression work required to reduce the molar volume of the natural gas by 20%. Report the work in units of hp.

From Appendix B, the critical temperature, pressure, and acentric factor for natural gas are T_C = 359.23 °R, and P_C = 670.13 psia, and ω = 0.0189.

Solution, part (a):

$$\frac{56,219,000\,ft^3}{day}\cdot\frac{1\,day}{24\,h}\cdot\frac{1\,h}{3600\,s}\cdot\frac{1\,lbmol}{376.42\,ft^3}=\underline{\underline{1.7286}}\ \frac{lbmol}{s}$$

ANS

Solution, part (b):

```
T = 519.67; (*Rankine, given*)
R = 10.73; (*ft^3*psia/(lbmol*R), given*)
ω = 0.0189; (*App. B, p.665*)
Tc = 359.23; (*Rankine, given*)
Pc = 670.13; (*psia, given*)
Tr1 = T / Tc; (*reduced temperature*)
(*Table 3.1, p.100*)
σ = 1;
ϵ = 0;
Ω = 0.08664;
Ψ = 0.42748;
α = (1 + (0.480 + 1.574 * ω - 0.176 * ω^2) * (1 - √Tr1))^2;
(*Equations 3.41, 3.44, and 3.45*)
b = Ω * R * Tc / Pc;
a = Ψ * α * R^2 * Tc^2 / Pc;
P = R * T / (V - b) - a / ((V + ϵ * b) * (V + σ * b));
```

$$W = -\int_{376.42}^{0.8*376.42} P\,dV \quad (*\text{eq 1.4, units } \frac{psia*ft^3}{lbmol}*)$$

1 241.057

$$\frac{W * psia * ft^3}{lbmol} * \frac{\frac{lbf}{in^2}}{psia} * \left(\frac{12\ in}{ft}\right)^2 * \frac{1.728\,609\ lbmol}{s} * \frac{1.34102\ hp}{737.562\ \frac{ft*lbf}{s}}$$

561.6782 hp

ANS

Cadet: **SOLUTION**

Problem:	Weight:
C	80

0.500 kmol of an ideal gas in a closed system is changed from an initial state of 7.0 m^3, 3.0 bar and 505.2 K to a final state of 3.0 m^3, 7.0 bar, and 505.2 K by a two-step process consisting of an adiabatic compression followed by cooling at constant pressure.

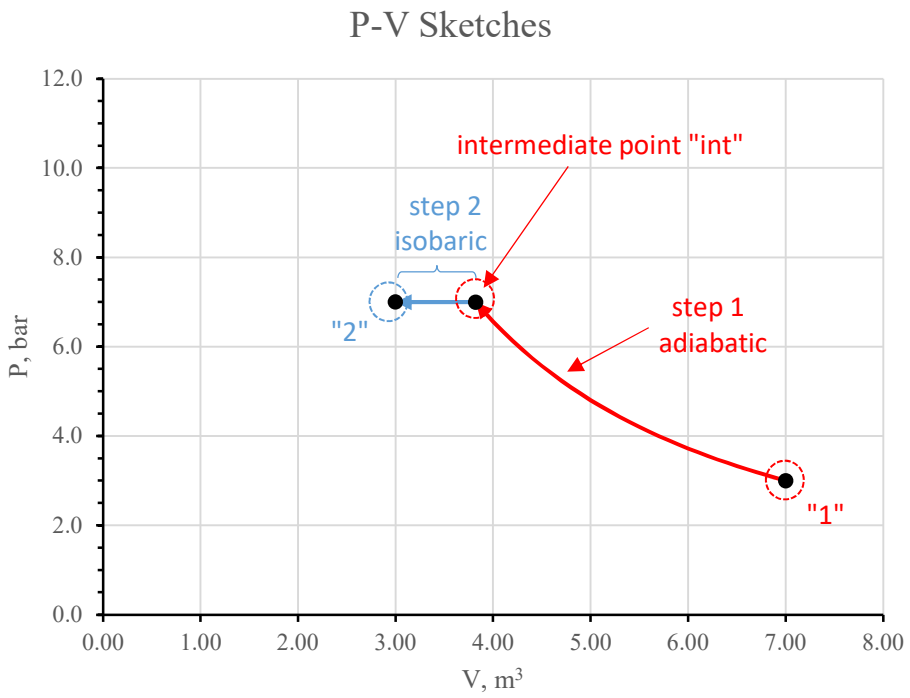
(a) Sketch the process path on the PV axes below.

(b) Calculate Q, W, ΔU and ΔH for the overall process in units of kJ.

(c) Calculate the intermediate temperature after step 1.

$$R = 8.314\ J/(mol\cdot K) = 0.08314\ (bar\cdot m^3)/(kmol\cdot K)$$
$$C_P = (7/2)\cdot R, \text{ and}$$
$$C_V = (5/2)\cdot R$$

Solution, part (a):



Solution, part (b):

The overall process is isothermal (T2=T1), so $\underline{\Delta H_{Overall} = \Delta U_{Overall} = 0\,kJ}$ ANS

Intermediate Volume (V_{int}):

Use equation 3.23c, with the intermediate pressure $P_{int} = P_2$ (Step 2 at constant P).

Equation 3.23c needs $\gamma = C_P/C_V = (7R/2)/(5R/2) = (7/5) = 1.4$ from equation 3.24.

$$P_{int}V_{int}^{\gamma} = P_1V_1^{\gamma} \Rightarrow V_{int} = \sqrt[\gamma]{\frac{P_1}{P_{int}}V_1^{\gamma}} = \sqrt[1.4]{\frac{3.0\,\text{bar}}{7.0\,\text{bar}}\left(7.0\,\text{m}^3\right)^{1.4}} = 3.821708\,\text{m}^3$$

Work:

Step 1, adiabatic, equation 3.26:

$$W_{Step1} = \frac{P_{int}\cdot V_{int} - P_1\cdot V_1}{\gamma - 1} = \frac{7.0\,\text{bar}\cdot 3.8217081\,\text{m}^3 - 7.0\,\text{bar}\cdot 3.0\,\text{m}^3}{1.4-1} = +14.37989\,\text{bar}\cdot\text{m}^3$$

Step 2, constant pressure, work = -PΔV:

$$W_{Step2} = -P\cdot\Delta V = -P\cdot\left(V_2 - V_{int}\right) = -7.0\cdot\left(3.0\,\text{m}^3 - 3.821708\,\text{m}^3\right) = +5.751956\,\text{bar}\cdot\text{m}^3.$$

Overall:

$$W_{Overall} = W_{Step1} + W_{Step2} = 14.37989\,\text{bar}\cdot\text{m}^3 + 5.751956\,\text{bar}\cdot\text{m}^3 = 20.13184\,\text{bar}\cdot\text{m}^3$$

Convert the units to kJ:

$$W_{Overall} = 20.1318\,\text{bar}\cdot\text{m}^3\left(\frac{1\,\text{J}}{10^{-5}\,\text{bar}\cdot\text{m}^3}\right)\cdot\left(\frac{1\,\text{kJ}}{1000\,\text{J}}\right) = \underline{\underline{2013.18\ \text{kJ}}}_{\text{ANS}}$$

Heat:

$$\Delta U_{Overall} = Q_{Overall} + W_{Overall} = 0 \Rightarrow Q_{Overall} = -W_{Overall} = \underline{\underline{-2013.18\,\text{kJ}}}_{\text{ANS}}$$

Solution, part (c):

$$PV = nRT \Rightarrow T_{int} = \frac{P_{int}\cdot V_{int}}{n\cdot R} = \frac{7.0\,\text{bar}\cdot 3.821708\,\text{m}^3}{0.500\,\text{kmol}\cdot\left(0.08314\,\frac{\text{bar}\cdot\text{m}^3}{\text{kmol}\cdot\text{K}}\right)} = \underline{\underline{643.54\,\text{K}}}_{\text{ANS}}$$